

Alpha decay 2

Let ν be the frequency of tunneling attempts. Then the decay rate λ is

$$\lambda = \nu e^{-2\gamma}$$

Gamow theorized that

$$\nu = \frac{v}{2r_1}$$

Velocity v is unknown but we can guess that v is proportional to $\sqrt{E/m}$ and write

$$\lambda = \frac{\beta \sqrt{E/m}}{2r_1} e^{-2\gamma}$$

Coefficient β is estimated to be

$$\beta = 843$$

The half-life $\tau_{1/2}$ is

$$\tau_{1/2} = \frac{\log 2}{\lambda} = \log 2 \frac{2r_1}{\beta \sqrt{E/m}} e^{2\gamma}$$

Hence

$$\tau_{1/2} = \frac{2 \log 2}{\beta} \frac{r_1}{\sqrt{E/m}} \exp \left(2C_1 \frac{Z}{\sqrt{E}} - 2C_2 \sqrt{Zr_1} \right)$$

For the alpha decay process ${}^{238}_{92}\text{U} \rightarrow {}^{234}_{90}\text{Th}$ we have

$$A = 238$$

$$Z = 90$$

$$E = 4.27 \text{ MeV}$$

$$r_1 = 7.44 \times 10^{-15} \text{ m}$$

Hence

$$\tau_{1/2} = 3.9 \times 10^{17} \text{ s} = 12 \times 10^9 \text{ yr}$$

The actual half-life of ${}^{238}\text{U}$ is

$$4.5 \times 10^9 \text{ yr}$$

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